# Chapter 4

# Nonlinear equations

## 4.1 Introduction

The pressure drop in a fluid in motion is examined. For a flow in a pipe with a circular cross section of diameter $D$ (meter), the Reynolds number, $Re$, is given by

$$Re = \frac{Dv}{\nu},$$

in which $v$ $(m/s)$ is the average flow velocity and $\nu$ $(m^2/s)$ is the viscosity of the fluid. The flow is called *laminar* if $Re < 2100$ (low flow velocity) and *turbulent* if $Re > 3000$. For $2100 \leq Re \leq 3000$, the flow is neither laminar nor turbulent.

For turbulent flows, the pressure drop between inflow and outflow is given by

$$P_{\text{out}} - P_{\text{in}} = \frac{\rho w L v^2}{2gD},$$

in which $w$ is a friction coefficient, $\rho$ $(kg/m^3)$ is the fluid density, $L$ $(m)$ is the length and $g$ $(m/s^2)$ is the acceleration of gravity. If the fluid contains particles (sand, paper fibers), then the friction coefficient $w$ satisfies the equation

$$\frac{1}{\sqrt{w}} = \frac{\ln(Re\sqrt{w}) + 14 - \frac{5.6}{k}}{k},$$

in which $k$ is a parameter known from experiments.

In this chapter, numerical methods will be discussed that can be used to determine $w$ if the values of $Re$ and $k$ are known.

## 4.2 Definitions

In this chapter, various iterative methods will be considered to solve nonlinear equations of the form $f(p) = 0$. The point $p$ is called a *zero* of the function $f$, or a *root* of the equation $f(x) = 0$. First, some useful definitions and concepts are introduced.

**Convergence**
Each numerical method generates a sequence $\{p_n\} = p_0, p_1, p_2, \ldots$ which should converge to $p$: $\lim_{n\to\infty} p_n = p$. Assume that the sequence indeed converges, with $p_n \neq p$ for all $n$. If there exist positive constants $\lambda$ and $\alpha$ satisfying

$$\lim_{n\to\infty} \frac{|p - p_{n+1}|}{|p - p_n|^{\alpha}} = \lambda, \tag{4.1}$$